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Unraveling the Lhasa-Qiangtang Collision in Western Tibet: Insights From Geochronological and Paleomagnetic Analyses

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Abstract The Lhasa-Qiangtang collision closed the Meso-Tethys Ocean, but the exact timing of this event remains hotly debated. Here, we present geochronological and paleomagnetic analyses conducted on Cretaceous volcanics from western Qiangtang to constrain the Lhasa-Qiangtang collision in western Tibet. Our investigations yield a paleolatitude of ~30.5 ± 5.0°N for western Qiangtang during ca. 110–100 Ma. A reanalysis of previously acquired Mesozoic-Cenozoic paleomagnetic data from western Qiangtang suggests a stationary position during ca. 136–34 Ma. Examination of paleomagnetic data from western Lhasa reveals a significant reduction in northward paleolatitudinal motion during the Early Cretaceous, dropping from ~12.3 cm/yr to nearly zero. Integration of our paleomagnetic findings with available geological records has led to conclude that the Lhasa-Qiangtang collision in western Tibet occurred at ca. 132 Ma. Additionally, we infer that crustal shortening on the order of ~1,000 km happened between Lhasa and Qiangtang during the Early Cenozoic.

Plain Language Summary The Tibetan Plateau comprises multiple different blocks, which originated from the Gondwana in the southern hemisphere. Their convergence histories toward Euraisa have changed the global land-sea distributions since the Late Paleozoic. The time at which the Lhasa block, one of the Tibetan blocks, accreted to the Qiangtang block to the north remains poorly constrained. In this work, we provide robust data suggesting a latitude of ~30.5 ± 5.0°N for western Qiangtang during the Early Cretaceous (ca. 110–100 Ma). We also compiled the available latitudinal data from western Tibet in combination with geological observations. We suggest Lhasa collided with Qiangtang during 132 million years ago in western Tibet. Significant shortening of the continental crust by ~1,000 km between the Lhasa and Qiangtang blocks occurred after their collision.

Key Points:

- Western Qiangtang had a paleolatitude of ~30.5 ± 5.0°N at ca. 110–100 Ma
- A substantial decrease in the paleolatitudinal motion of western Lhasa occurred in the Early Cretaceous
- The Lhasa-Qiangtang collision in western Tibet occurred at ca. 132 Ma

Supporting Information:

Supporting Information may be found in the online version of this article.

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1. Introduction

The evolution of the Tethyan domain during the formation and break-up of Pangea involved various stages of the Wilson cycle spanning late Paleozoic-early Cenozoic eras, characterized by the opening, growth, and closure of multiple oceans (i.e., the Paleo-, Meso-, and Neo-Tethys oceans) (L. Ding et al., 2022; A. Yin & Harrison, 2000). The Meso-Tethys ocean opened concurrently with the separation of the Cimmerian blocks (e.g., Qiangtang, Sibumasu) from southernly Gondwana (DeCelles et al., 2014; Metcalfe, 2013), and was then consumed by oceanic subduction beneath Qiangtang and/or Lhasa, accompanied by the rifting and northward drift of Lhasa toward northernly Eurasia (Kapp & Decelles, 2019). The Meso-Tethys eventually closed during the Lhasa-Qiangtang collision processes (Hu et al., 2022; S. Li et al., 2019), with remnants trapped in the Bangong-Nujiang suture zone (BNS) (Figure 1a and Figure S1 in Supporting Information S1). While central Tibet preserves extensive geological records related to these processes, many fundamental issues regarding the temporal-spatial evolution of the Meso-Tethys remain contentious. The rifting of Lhasa from either Indian (Torsvik & Cocks, 2013) or Australian Gondwana (Q. Wang et al., 2021) is interpreted to have occurred either during the Early Middle Permian (marked by emplacement of significant volumes of mafic rocks, Liao et al., 2015), or during the Middle-Late Triassic (as suggested by paleolatitudinal reconstructions, Z. Li et al., 2016). The subduction of the Meso-Tethys oceanic crust exhibited either a single polarity beneath Lhasa (Zhu et al., 2009) or Qiangtang (Y. Li et al., 2013), or a divergent double polarity beneath both (Kapp & Decelles, 2019) during the Middle Permian-Early Jurassic. The Lhasa-Qiangtang collision is suggested to have occurred from the Middle-Late Jurassic to the Late Cretaceous (Guynn et al., 2006; Kapp et al., 2007; S. Li et al., 2019; Z. Yin et al., 2024).

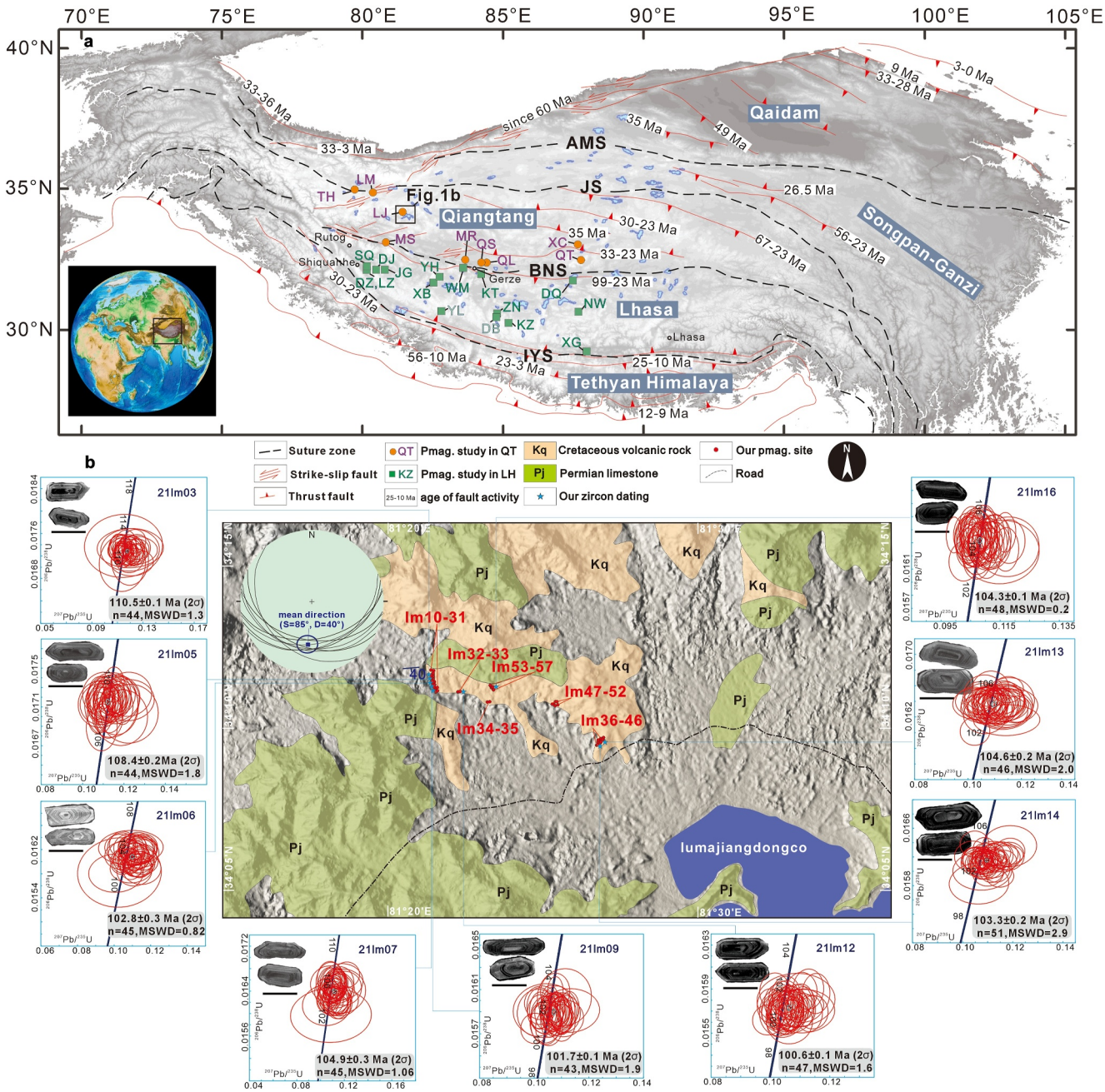


Figure 1. Illustration of the tectonics of Tibet and sampled area. (a) Tectonic and topographic map of the Tibet. Blue dots/green squares show sampling locations of previous Mesozoic-Cenozoic paleomagnetic data in western Qiangtang (QT) and Lhasa (LH), respectively. Abbreviations: QT-Lippert et al., 2011; MS-this study; QL, QS -W. Chen et al., 2017; MR-Cao et al., 2020; LM-Y. Chen et al., 1993; TH-Y. Chen et al., 1993; XC-C. Song et al., 2012; DQ-Meng et al., 2017; KT-J. Ding et al., 2015; DZ-Y. Ma et al., 2017; LZ, YL-Yi et al., 2015; KZ-X. D. Tang et al., 2013; JG-Y. Ma et al., 2018; NM-Niu et al., 2023; WM-S. Wang et al., 2022; DJ-Bian et al., 2017; ZN-W. Chen et al., 2012; Yang et al., 2015; YH-Y. Ma et al., 2014; SQ-Y. Chen et al., 1993; XB-Z. Li et al., 2022; XG-Y. Ma et al., 2022; DB-Zhou et al., 2016 (Table S5 in Supporting Information S1). Pmag.-paleomagnetism. AMS-Ayimaqin suture; JS-Jinsha suture; BNS-Bangong-Nujiang suture; IYS-Indo-Yarlung-Zangbo suture. (b) Simplified geological map of sampled Cretaceous lavas in western Qiangtang with zircon U/Pb dating data of nine volcanic rocks. Black scale represents 100 μm.

Specifically, a scissor-like Lhasa-Qiangtang collision process was proposed to have occurred between ~162 Ma and the mid-Cretaceous, with initiation in the east and progressive movement to the west (e.g., Kapp et al., 2007; Liu et al., 2017), consistent with the rotation patterns of Qiangtang and Lhasa (Yan et al., 2016). Conversely, others argue for a synchronous collision during the Cretaceous (e.g., Y. F. Chen et al., 2020; K. J. Zhang et al., 2012).

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These controversies have resulted in diverse tectonic models of the Meso-Tethys, complicating our understanding of Tethyan tectonic evolution and global land-sea distributions. Paleomagnetism aids in reconstructing latitudinal plate and block motion (e.g., Huang et al., 2018), therefore Mesozoic paleomagnetic data are crucial for understanding the convergence history of Tibetan blocks. Some suggest western Qiangtang accreted to the Eurasian margin in the Late Triassic (C. Song et al., 2012), and experienced negligible latitudinal motion (~28–30°N) from 120 to 30 Ma (W. Chen et al., 2017; Cao et al., 2020; Lippert et al., 2011; P. Song et al., 2022). However, others argue for a more southerly latitude of ~21–22°N, proposing a pre-120 Ma Lhasa-Qiangtang collision in western Tibet (Bian et al., 2017; S. Wang et al., 2022). Here, we present new geochronological and paleomagnetic data from Lower Cretaceous lavas from western Qiangtang to determine the paleolatitude, and integrate these findings with available paleomagnetic and geological data to evaluate the timing of the Lhasa-Qiangtang collision in western Tibet.

2. Geological Setting and Sampling

Tibet is subdivided into Tethyan Himalaya, Lhasa, Qiangtang, and Songpan-Ganzi blocks, separated by the Indus-Yarlung-Zangbo, Bangong-Nujiang, and Jinsha sutures, respectively (Figure 1 and Figure S1 in Supporting Information S1). Some suggest the large metamorphic belt in the Longmu Co-Shuanghu region represents an in situ ophiolite zone separating western and eastern Qiangtang subblocks (e.g., Zhai et al., 2013). Others argue it formed due to thrusting beneath Qiangtang from the Jinsha suture, indicating an intra-continental setting (e.g., Pullen & Kapp, 2014). Upper Paleozoic strata containing glacio-marine sediments crop out extensively in western Qiangtang (K. J. Zhang et al., 2012). Upper Triassic to Middle Jurassic marine sediments unconformably overlie these strata (A. Ma et al., 2017). In the Early Cretaceous, the marine environment transitioned into a continental environment throughout western Qiangtang (Kapp et al., 2007). Magmatism in western Qiangtang is characterized by three main phases: Carboniferous-Permian rift-related (Y.-X. Zhang & Zhang, 2016), Late Jurassic-Cretaceous syn-collision (Y. Li et al., 2018; Liu et al., 2017), and Eocene-Oligocene post-collision phases (L. Ding et al., 2007).

The Cretaceous volcanics we investigated in western Qiangtang are mainly gray and dark andesites conformably overlying sandstones (Figures S2 and S5 in Supporting Information S1). We drilled 48 volcanic sites from monoclinically folded strata near Lumajiangdongco (34.2°N, 81.4°E), using a portable gasoline-powered driller, with orientations determined using both magnetic and sun compasses. At each site, 8–10 paleomagnetic samples were collected, spanning several meters of stratigraphic thickness and/or at least one lava flow. Approximately 25 lava flows were sampled across ~1,500 m. Bedding attitudes were assessed from sandstones and volcanic cooling margins (Figure S2 in Supporting Information S1), with an average dip of 40° toward 175° from nine directional measurements conducted in the field (Figure 1b). See Supporting Information S1 for details on the U/Pb dating, rock magnetism, magnetic petrography, and paleomagnetism analysis methods and the data.

3. Results

3.1. Geochronology

Nine geochronological samples were collected to determine the eruption timing of the lavas using zircon U/Pb laser-ablation-multicollector inductively coupled plasma mass spectrometry (LA-ICP-MS) methods. The inferred crystallization time of a single sample was derived from the mean age of its youngest group of zircons. Cathodoluminescence images of individual crystals from these samples exhibit euhedral shapes without xenocrystic cores (Figure 1b). The Th/U rations of the youngest zircon population of all nine samples ranged from 0.87 to 7.63, with an average mean of 2.78, showing a magmatic origin (see Table S1 in Supporting Information S1). In the stratigraphic sequence, samples 21lm03, 21lm05, 21lm06, 21lm07, and 21lm09 were collected from sites lm10-33, yielding $^{206}Pb/^{238}U$ weighted mean ages of 110.5 ± 0.1 Ma (±2σ, n = 44, MSWD = 1.3), 108.4 ± 0.2 Ma (±2σ, n = 44, MSWD = 1.8), 102.8 ± 0.3 Ma (±2σ, n = 45, MSWD = 0.82), 104.9 ± 0.3 Ma (±2σ, n = 45, MSWD = 1.06), and 101.7 ± 0.1 Ma (±2σ, n = 43, MSWD = 1.9) (Figure 1b), respectively. Samples 21lm12, 21lm13-14, and 21lm16 were collected from sites lm32–33, lm36-46, and lm53-57, respectively. These samples were dated to $^{206}Pb/^{238}U$ weighted mean ages of 100.6 ± 0.1 Ma (±2σ, n = 47, MSWD = 1.6), 104.6 ± 0.2 Ma (±2σ, n = 46, MSWD = 2.0), 103.3 ± 0.2 Ma (±2σ, n = 51, MSWD = 2.9), and 104.3 ± 0.1 Ma (±2σ, n = 48, MSWD = 0.2). These nine zircon U-Pb ages indicate that the sampled volcanic rocks have an Albian age (~100–110 Ma), contradicting a Late Carboniferous-Early Permian age assigned by the

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1:250,000 scale Tuozegangri regional geological survey report (144C002003). It is worth noting that Cretaceous volcanism (ca. 120–90 Ma) was prolonged in the Gerze-Seilingco area of Qiangtang (Kapp et al., 2005; Liu et al., 2018).

3.2. Paleomagnetism

The intensity of the natural remanent magnetization (NRM) typically falls within the range of 0.2–20 mA/m, with a minor occurrence of a higher value at ∼5–20 A/m. Thermal demagnetization treatments successfully isolated two distinct magnetic components in one-in-three specimens (Figure 2a). These, referred to as low-temperature (LT) and high-temperature (HT) components, were generally removed below 300°C and within temperature intervals between ∼300°C and 580–680°C, respectively (Figure 2). The mean direction of the LT is D = 4.7°, I = 49.1°, with α_{95} = 5.4° (n = 86 samples, in situ coordinates), which closely resembles the International Geomagnetic Reference Field (IGRF) direction (D = 1.8° and I = 53.4°) in this region, indicating that the LTs likely represent viscous remanent magnetizations acquired recently (Figure 2b). The HT directions from sites lm11, lm12, lm13, lm35, lm47, and lm52 were discarded from further analysis due to the failure of our data filters introduced in Supporting Information S1. Analysis of the residual remanence intensity above 580°C relative to NRM intensity (Int_{580}/Int_{NRM}) across the 281 samples from 42 remaining sites reveals that ∼90% of the samples (254 samples) have ratios below 0.4, with a peak value of 0.12 (Table S5 and Figure S8 in Supporting Information S1). Only 25 samples from sites lm53-57 exhibit Int_{580}/Int_{NRM} ratios greater than 0.5. Considering the saturation magnetization of maghemite is similar to that of magnetite, the abundance of magnetite in our samples is inferred to be much higher than that of maghemite, with a low degree of oxidation, as supported by our rock magnetism and magnetic petrographic observations (see Supporting Information S1). As maghemite with low and medium degrees of oxidation is expected to preserve the original natural remanence magnetization of magnetite (Doubrovine & Tarduno, 2004; Johnson & Merrill, 1974), our HT component defined within the temperature range of ∼350 and ∼680°C likely represent the primary magnetization directions carried by (weakly Ti-substituted) magnetites during initial cooling. Indeed, the directions carried by magnetite during the thermal demagnetization level of ∼350–580°C closely match those carried by maghemite between 580°C and ∼680 (Figure 2a). The HTs exclusively exhibit normal geomagnetic polarities directed to the east and down (Figure 2b). Based on these observations, we calculate a mean pole at 43.9°N, 172.5°E with A_{95} = 3.5° in in situ coordinates, yielding a corresponding in situ mean inclination of 39.3 ± 3.9° and declination of 51.1 ± 3.0° (N = 42 sites). The Fisher mean (Fisher, 1953) virtual geomagnetic pole (VGP) in tilt-corrected coordinates is situated at 15.4°N, 144.7°E with A_{95} = 5.0°, resulting in a mean direction of 49.7 ± 4.5° inclination and 91.6 ± 4.6° declination (Figure 2; Tables S2 and S3 in Supporting Information S1).

3.3. Reliability of the Characteristic Directions

First, we employed a fold test to compare our paleomagnetic data with that from ∼104 Ma Qushenla Formation lavas in Gerze (W. Chen et al., 2017), as our sampled strata exhibit monoclinic folding. Our tilt-corrected mean inclination (I = 49.7°) matches that of W. Chen et al. (2017) (I = 50.66), but a ∼100° difference in declination suggests regional vertical-axis rotation between Lumajiangdongco and Gerze (Figure 2b), supported by fault investigations (Taylor et al., 2003; Figure 1a). We calculated site-mean inclinations for our studied area (34.2°N, 81.4°E) using the corresponding VGPs from W. Chen et al. (2017). After rotating W. Chen et al.'s site mean declinations by 100° clockwise both in geographic and stratigraphic coordinates (see Table S3 in Supporting Information S1), the progressive unfolding correction of all 56 sites yielded a positive result with a highest precision parameter at 95.2% (Figure 2c). The tilt-corrected *k* increased significantly from 11.2 to 31.2. The Fisher-mean inclination (I = 50.0°) matches the inclination-only mean (I = 49.6 ± 3.1°) using the method of Arason and Levi (2010). Second, the HTs are primarily carried by single-domain and multi-domain magnetites, with minor maghemite and hematite (see Supporting Information S1). Third, the HTs align with the Cretaceous Normal Polarity Superchron (126–84 Ma, Gradstein et al., 2012). No secondary magnetizations were found for the HT components, supporting a primary origin (see Supporting Information S1). Furthermore, the angular dispersion (S) value (18.2°) of the 42 site-mean VGPs matches the PSV10 data set at a similar latitude (de Oliveira et al., 2021). The A_{95} value of 5.0° of mean pole falls within the expected interval (2.69°, 7.78°) for the paleosecular variation (PSV) of 42 poles (Deenen et al., 2011). This reasonable PSV suggests our data set may have averaged secular variation, thus providing a statistically rigorous estimate of the time-averaged field during the eruption of the lavas. Consequently, based on our findings, we conclude that the ChRM directions from our 42

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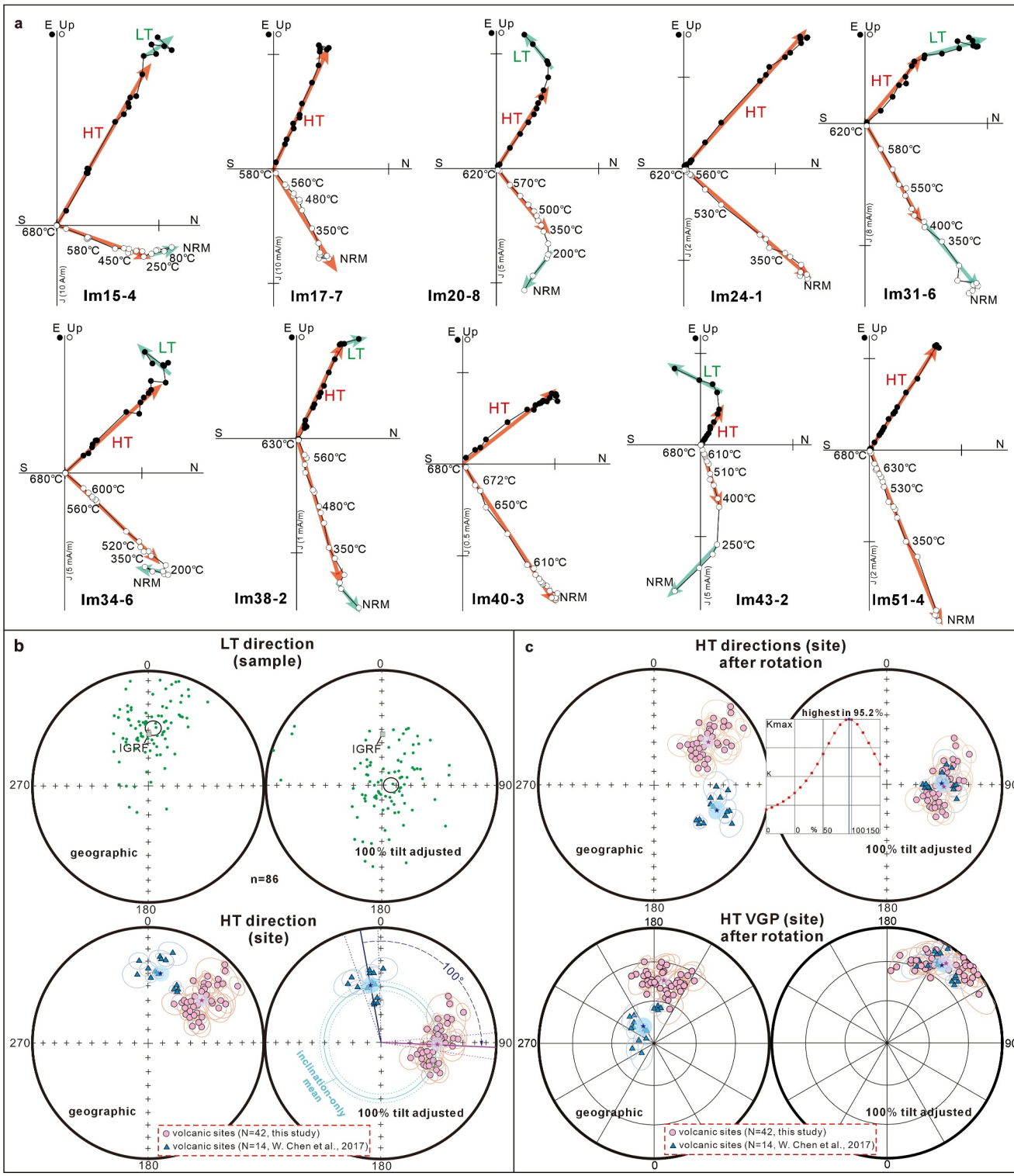


Figure 2. Interpretable paleomagnetic results of Cretaceous lavas. (a) Orthogonal vector diagrams of thermal demagnetization behaviors of typical volcanic specimens in in situ coordinates, showing low-temperature (LT) and high-temperature (HT) components. (b) Equal-area projections of the paleomagnetic sample-mean low-temperature (LT) and site-mean high-temperature (HT) component directions of this study and W. Chen et al. (2017). (c) HTC and VGP directions of this study and W. Chen et al. (2017) after ca. 100° clockwise rotation and progressive fold test.

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sites indeed record a primary origin, yielding a latitude of 30.5 ± 5.0°N for a reference point (34.2°N, 81.4°E) in western Qiangtang at ~110–100 Ma.

4. Discussion

Our discussion in this paper focuses on the convergence history between Qiangtang-Lhasa in western Tibet (longitudinally from 80°E to 90°E). A summary of available Mesozoic-Cenozoic paleomagnetic results is presented in Table S6 of the Supporting Information S1 and Figure 3a. Paleolatitudes have been calculated for a reference point in our sampling area (34.2°N, 81.4°E).

4.1. Mesozoic-Cenozoic Paleolatitudes of Western Qiangtang

Paleomagnetic data derived from Xiaochaka (XC) sandstones suggest that western Qiangtang accreted to Eurasia at 25.6 ± 3.8°N in the Late Triassic (C. Song et al., 2012), coinciding with the closure of the Paleo-Tethys ocean. However, the interpretation of five available paleomagnetic paleopoles, obtained from Cretaceous sandstones and volcanic rocks show considerable disparity (Cao et al., 2020; W. Chen et al., 2017; Y. Chen et al., 1993). The paleolatitudes derived from the Lower Cretaceous Longmucuo (LM) and Tianshuihai (TH) and the Upper Cretaceous Qushenla (QL) redbeds range from 10.6 ± 5.1°N (Y. Chen et al., 1993) to 31.8 ± 3.2°N (W. Chen et al., 2017). In contrast, the paleolatitudes from the ~104 Ma Qushenla (QS) and ~116 Ma Meriqieco (MR) volcanic rocks yield values of 31.5 ± 5.7°N (W. Chen et al., 2017) and 26.4 ± 5.0°N (Cao et al., 2020), respectively. The discrepancy may be contributed to the following factors: (a) Sedimentary paleomagnetic data, such as those reported by Y. Chen et al. (1993), may exhibit lower latitudes due to inclination shallowing, and (b) while volcanic rocks are immune to the inclination shallowing bias, pelaomagnetic data from the QS and MR volcanic rocks may not sample enough individual VGP readings to average out paleomagnetic virtual secular. S. Wang et al. (2022) used small circle and inclination-only methods on all five paleopoles derived from both sedimentary and volcanic rocks, obtaining paleolatitudes of 18.9 ± 11.8°N and 21.4 ± 2.9°N, respectively, including results from the LM and TH redbeds without inclination shallowing correction. Given that the prevalent sedimentary inclination shallowing observed throughout Central Asia (e.g., Lippert et al., 2011; Yan et al., 2005), the Late Cretaceous QL paleomagnetic results from redbeds were subjected to elongation/inclination analysis (Tauxe & Kent, 2004), with corrected inclinations used for subsequent study. Considering that the local vertical-axis rotations within Tibet occurred along faults (Figure 1a; Taylor et al., 2003), and that more sites or independent readings sampling the time-varying paleomagnetic field can significantly improve the accuracy of paleolatitude estimation (Gerritsen et al., 2022; Sapienza et al., 2023), we applied an inclination-only method to paleomagnetic inclinations from the LJ, QS, and MR volcanic rocks and corrected inclinations from the QL redbeds, recalculated from site VGPs to the reference point, following the method of Arason and Levi (2010). The 96 Cretaceous inclinations collectively yield a mean of 49.01 ± 2.09 (α_{95})° (mean-M1), corresponding to a paleolatitude of 29.9 [28.1, 31.8]°N (see Table S7 in Supporting Information S1). Additionally, two Eocene paleopoles from central Qiangtang (QT) and Meisu (MS) volcanic rocks indicate paleolatitudes of 29.9 ± 3.7°N (Lippert et al., 2011) and 30.2 ± 5.9°N (P. Song et al., 2022), respectively. These results collectively indicate western Qiangtang experienced no significant latitudinal movement from the Early Cretaceous to the Late Eocene (Figure 3a).

4.2. Mesozoic-Cenozoic Paleolatitudes of Western Lhasa

For western Lhasa, only three Triassic-Jurassic paleomagnetic studies have been published (Z. Li et al., 2022; Y. Ma et al., 2022; Zhou et al., 2016). One paleomagnetic study on the Late Triassic Dibuco (DB) limestone yielded a latitude of 18.4 ± 10.7°S (Zhou et al., 2016). Two paleopoles derived from the ~173–183 Ma Bima (XG) and ~153–156 Ma Zenong (XB) volcanic rocks indicate paleolatitudes of 6.9 ± 5.0°S (Y. Ma et al., 2022) and 4.6 ± 2.5°S (Z. Li et al., 2022), respectively. We note, that a "monster shift" true polar wander (TPW) has been proposed to have occurred in Late Jurassic-earliest Cretaceous (ca. 160–145 Ma) (e.g., Fu & Kent, 2018; Kent & Irving, 2010; Muttoni et al., 2005), characterized by a ~30–20° clockwise rotation with an Euler axis at 0°N, 11°E and a velocity of ~2°/Myr. Western Lhasa, located to the east of the Euler pole, could have been translated to a more southernly latitude by this potential TPW. Therefore the Jurassic paleolatitude might reflect the interplay between the block's northward plate tectonic motion and the simultaneous TPW (Z. Li et al., 2022). However, the existence and speed of this TPW event remain hotly debated and depend on different data filters. A reassessment of the Eurasian and Indian paleopoles with careful data selection ruled out this anomalously rapid TPW (Kulakov

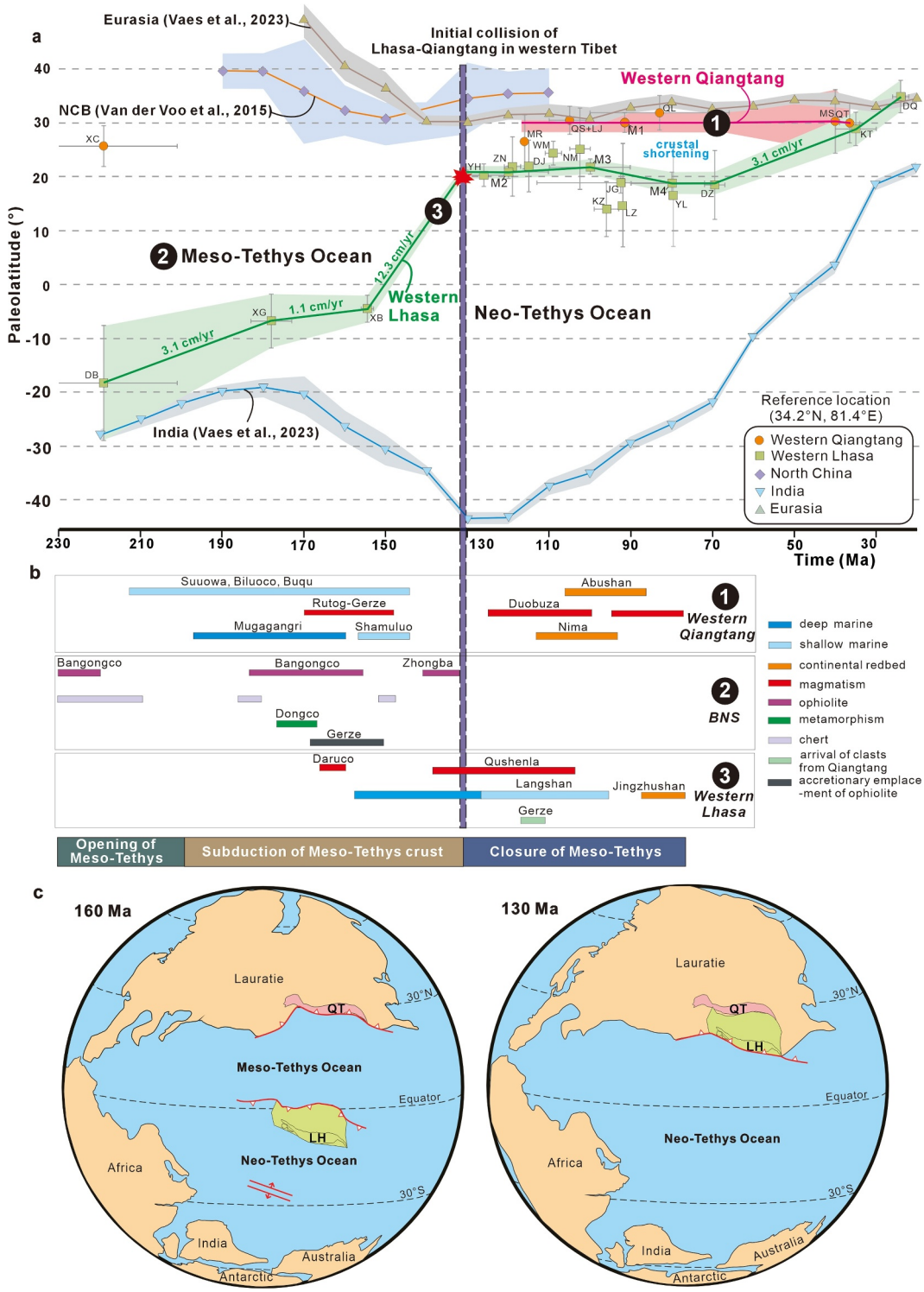


Figure 3. Paleomagnetic and geological evidences for evolutions of the Meso-Tethys Ocean and paleogeographic reconstructions. (a) Paleolatitude versus time plot showing paleomagnetic data from western Tibet (Table S5 in Supporting Information S1). Abbreviations refer to Figure 1b and associated references. Eurasian and Gondwanan-India paleolatitudes are from Vaes et al. (2023); North China from Van der Voo et al. (2015). (b) Summary of ophiolite occurrences, lithostratigraphic, magmatic, and tectonic events in western Tibet, modified from Hu et al. (2022). (c) Paleogeographic reconstructions showing Qiangtang (QT) and Lhasa (LH) at ca. 160 and 130 Ma, within the global geography of Müller et al. (2019).

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et al., 2021; Torsvik et al., 2012; Vaes et al., 2023), suggesting instead the possibility of steady slow continental rotation at a rate of <0.8°/Myr throughout the Jurassic. Assuming a constant paleolatitudinal drift rate (~3.1 cm/yr) for western Lhasa during ca. 180–155 Ma (Figure 3a), we predict a paleolatitude of ~0.1°N for the ca. 155 Ma XB lavas. Under this assumption, the potential TPW could have produce only a ~4.7° southernly displacement of the XB latitude in western Lhasa. Additionally, Hou et al. (2024) suggest that the onset of the Jurassic TPW occurred at ca. 155 Ma; therefore, the impact of the potential TPW on the paleolatitudinal displacement of ca. 156–153 Ma XB volcanic rocks may have been limited. These suggest western Lhasa remained positioned within the tropical-subtropical zone of the southern hemisphere until the middle Late Jurassic (Z. Li et al., 2022). Significant south-to-north motion (~4,000 km) of Lhasa toward the northern hemisphere is inferred to have happened since ~153 Ma.

Numerous Cretaceous paleomagnetic studies have been conducted in western Lhasa, providing good constrains on its Cretaceous latitude. Four paleopoles obtained from the Shiquanhe limestone (SQ), Jiega limestone (DJ), Jiega sandstone (JG), and Risong (WM) redbeds yield latitudes of 12.9 ± 13.1°N (Y. Chen et al., 1993), 21.9 ± 4.8°N (Bian et al., 2017), 18.8 ± 1.9°N (Y. Ma et al., 2018), and 24.3 ± 2.2°N (S. Wang et al., 2022), respectively. Since sedimentary rocks may undergo inclination shallowing and the SQ result lacks inclination corrections, it is excluded from further discussion. The DJ, JG, and WM sedimentary results, however, are considered reliable and are utilized, as they have been proven immune of inclination shallowing (Bian et al., 2017; Y. Ma et al., 2018; S. Wang et al., 2022). Additionally, six paleopoles derived from the DZ, LZ, YL, KZ Linzizong, Zenong (ZN), and Qushenla (YH, NM) volcanic rocks yield paleolatitudes of 18.4 ± 6.4°N (Y. Ma et al., 2017), 16.5 ± 9.6°N (Yi et al., 2015), 14.6 ± 2.7°N (Yi et al., 2015), 13.9 ± 5.1°N (X. D. Tang et al., 2013), 21.8 ± 5.5°N (W. Chen et al., 2012; Yang et al., 2015), 20.2 ± 2.1°N (Y. Ma et al., 2014), and 25.2 ± 7.5°N (Niu et al., 2023), respectively. These collective Cretaceous data suggest that Lhasa has been in the northern hemisphere since ~132 Ma. Given the abundant paleomagnetic data spanning ~132–67 Ma and the prevalent local rotations in western Lhasa, we employed a ~20 Myr window to average the inclination-only data recalculated from site VGPs to the reference point using the Arason and Levi (2010) method (see Table S8 in Supporting Information S1). Three mean-inclinations of 36.8 ± 1.7 (α_{95})°(mean-M2 from YH (51 sites), ZN (30 sites), and DJ (25 sites)), 38.3 ± 2.0° (mean-M3 from WM (56 sites), NM (24 sites), KZ (10 sites), JG (38 sites), and LZ (10 sites)), and 33.8 ± 3.1°(mean-M4 from DZ (44 sites) and YL (15 sites)) are obtained during 132–110, 110–90, and 90–67 Ma, respectively (see Table S8 in Supporting Information S1). Mean-M2, -M3, and -M4 correspond to paleolatitudes of 20.5 [19.4, 21.7]°N, 21.5 [20.2, 23.0]°N, and 18.5 [16.5, 20.6]°N, respectively (see Table S8 in Supporting Information S1). Considering the consistency of these three paleolatitude estimates within their uncertainties, we suggest that western Lhasa did not experience detectable paleomagnetic displacement during 132–67 Ma (Figure 3a). Moreover, Eocene (KT) and Oligocene (DQ) paleopoles derived from shallowing-corrected sedimentary inclinations reveal a significant northward displacement of ~1,700 km during ca. 67–22 Ma.

4.3. Implications for the Lhasa-Qiangtang Collision in Western Tibet

Our compilation of paleomagnetic data from western Qiangtang indicates that it was situated at ~29.9°N and underwent negligible latitudinal movement during the period of ca. 116–34 Ma. Furthermore, our findings suggest that western Lhasa began to rift away from Gondwana during the Late Triassic to Early Jurassic, coinciding with the extension of the Meso-Tethys Ocean (Figure 3b). The maximum width of the Meso-Tethys Ocean is estimated to have been ~4,000–5,000 km. Western Lhasa exhibited a northward drift at a velocity of ~3.1 cm/yr from 220 to 180 Ma and ~1.1 cm/yr from 180 to 155 Ma, followed by a rapid increase to ~12.3 cm/yr until Lhasa reached the northern hemisphere at ~132 Ma. We note that, the Jurassic paleolatitudinal motion of western Lhasa could be slightly biased by the potential TPW (see Section 4.2). Since then, western Lhasa has remained stationary at ~18.5–21.5°N for ~50 Myr (Figure 3a).

We then conducted an integrated analysis of ophiolite occurrences, lithostratigraphy, magmatism, and tectonic events in western Tibet to constrain the Lhasa-Qiangtang collision. Ophiolites in the western BNS were dated primarily to ca. 193 and 147 Ma in the Bangongco and Zhongba areas (e.g., Hu et al., 2022; S. Li et al., 2019; Shi, 2007; Y. Tang et al., 2020; B. D. Wang et al., 2016), with a youngest age of ~133 Ma, suggesting ongoing formation of oceanic crust possibly until the late Early Cretaceous. The accretionary emplacement of the Gerze ophiolites occurred during ~151–168 Ma (S. Li et al., 2019), preceding the collision. The marine environment in western Qiangtang started during the Late Triassic-Early Jurassic and was ceased by the Late Jurassic, with

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marine rocks unconformably overlain by Upper Cretaceous continental red beds (A. Ma et al., 2017). In contrast, the marine environments in western Lhasa continued to develop until Mid-Cretaceous (Rao et al., 2020). The arrival of Qiangtang detritus into western Lhasa basins at ~111–116 Ma marks a minimum age for the initial Lhasa-Qiangtang collision (S. Li et al., 2019). The ca. 131–110 Ma Qushenla volcanics in western Lhasa record first roll-back and then break-off of south-dipping Meso-Tethys lithosphere following the collision (e.g., S.-M. Li et al., 2018). The scarcity of magmatic records from ca. 142 to 128 Ma in western Qiangtang has been interpreted to reflect the termination of subduction and therefore the initial collision (e.g., Kapp et al., 2007; Zhu et al., 2016). Overall, these geological observations indicate that the Lhasa-Qiangtang collision likely occurred during ca. 133–116 Ma (Figure 3b). Our paleomagnetic analysis reveals that the Cretaceous standstill of western Lhasa following rapid northward motion happened after ~132 Ma. In summary, the combination analysis of paleomagnetic data set and geological records support a ca. 132–133 Ma timing for the Lhasa-Qiangtang collision in western Tibet (Figure 3c). Future research focusing on high-quality paleopoles from Upper Jurassic-Lowest Cretaceous units both in western Lhasa and Qiangtang will further refine the processes involved in the evolution of the Meso-Tethys.

The comparison of the calculated paleolatitudes between western Lhasa and Qiangtang suggests a ~9.3° discrepancy in latitude in 132–67 Ma. Instead of a spatial gap interpretation, we use the crustal shortening model of P. Song et al. (2022) to explain this. The latitudinal velocity of western Lhasa was ~3.1 cm/yr during ca. 67–22 Ma, causing significant crustal shortening (~1,000 km) within the western Lhasa and/or BNS during the early stage of the India-Asia collision. This estimate is approximately 500–600 km greater than estimates based on geological records (Murphy et al., 1997; Volkmer et al., 2007). This discrepancy may be due to either significant, previously undetected tectonic shortening within the interior of the Lhasa/BNS, or/and the eastward extrusion of the Indochina block, which originally occupied a position between Lhasa and Qiangtang (Dupont-Nivet et al., 2010; Otofuji et al., 2007).

5. Conclusions

New geochronological and paleomagnetic data obtained from Early Cretaceous volcanic rocks in western Qiangtang reveal a paleolatitude of 30.5 ± 5.0°N at ca. 110–100 Ma. A comparison of our new data and previous Mesozoic-Cenozoic paleomagnetic data from western Qiangtang and Lhasa provides key constraint on the convergence evolution of the two blocks. We conclude that western Qiangtang occupied a stationary position (~29.9°N) during the Cretaceous-Eocene period. The northward drift velocity of western Lhasa decreased significantly from ~12.3 cm/yr to standstill during the Early Cretaceous. Based on these findings integrated with geological records, we propose that the Lhasa-Qiangtang collision in western Tibet occurred at ca. 132 Ma.

Data Availability Statement

Data used in this article is available in P. Song (2024).

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